Supplementary Information
PhysBench: a verifiable multi-domain benchmark for physics-informed learning

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SI 1. Parameter ranges

Table 1 lists the full parameter ranges and the deterministic seeding protocol for each domain.

Table 1: PhysBench parameter ranges.

Domain	Parameters (range)
beam	$L \in [1, 6]$, $P \in [10^3, 5 \times 10^4]$, $E \in [69, 210]\,\mathrm{GPa}$, $I \in [10^{-7}, 2 \times 10^{-5}]$, $h \in [0.05, 0.4]$
cantilever	identical ranges to beam (identical token stream)
projectile	$v_0 \in [5, 30]\,\mathrm{m/s}$, $\alpha \in [15°, 75°]$
pendulum	$L \in [0.5, 3]$, $\theta_0 \in [5°, 60°]$
spring	$k \in [1, 50]$, $m \in [0.1, 5]$, $A \in [0.2, 2]$
rc	$R \in [100, 10^4]$, $C \in [10^{-6}, 10^{-3}]$, $V_0 \in [1, 12]$
burgers	$\nu \in [0.02, 0.1]$, $A \in [0.8, 1.8]$, $\sigma \in [0.25, 0.45]$
heat2d	mode numbers $k, l \in \{1, 2, 3\}$, amplitudes $\in [0.5, 2]$

SI 2. Solver and verifier independence

For the trajectory domains, the generator is the exact closed-form solution (`physx/sim.py`) and the verifier is a finite-difference discretization of the governing equation (`physx/residuals.py`): e.g. for the beam, the generator evaluates the closed-form deflection $w(x) = Px(4x^2 - 3L^2)/(48EI)$ (piecewise), while the verifier checks the fourth-derivative equilibrium $EI\, w'''' = -q$ numerically on the predicted curve. The two code paths share no implementation. For Burgers, the generator is a high-resolution upwind scheme; the verifier evaluates $u_t + uu_x - \nu u_{xx}$ by central differences on the predicted grid. For heat2d, the generator sums the separable-mode solution; the verifier evaluates $\nabla^2 u - f$ on the predicted grid.

SI 3. The baseline matrix, per-seed

The full 75-run matrix (per-seed training rel-MAE) is committed under `physx/models/matrix/`. Table 2 summarizes means and standard deviations per cell.

Table 2: Baseline matrix (mean ± std over 5 seeds).

Domain	phys	no-physics	MLP
beam	0.118 ± 0.029	0.153 ± 0.048	0.133 ± 0.037
cantilever	0.120 ± 0.027	0.150 ± 0.030	0.134 ± 0.030
burgers	0.008 ± 0.001	0.010 ± 0.002	0.016 ± 0.003
heat2d	0.014 ± 0.007	0.013 ± 0.003	0.020 ± 0.007
projectile	0.100 ± 0.031	0.057 ± 0.030	0.068 ± 0.025

SI 4. Reproducibility

```
python3 -c "from physx import dataset; print(len(dataset.generate('beam', n=256, seed=0)))"
python3 physx/run_matrix.py        # rerun the 75-run baseline matrix
python3 -m unittest physx.test_physx   # 49 tests, incl. dataset/verifier tests
```

Reproducing the matrix takes under an hour on a 20-core CPU; data generation is instantaneous and deterministic.